



Ebury What borders?®

Case Study

Senior Data Engineer (Platform)

Take home task

Objective

Develop a data pipeline using Airflow, dbt, and PostgreSQL to ingest, transform, and expose data. The pipeline should be containerized and deployable using Docker Compose. The goal is to demonstrate the candidate's proficiency in data engineering workflows.

Task Description

1. Data ingestion and preparation

- Use the provided dataset (`customer_transactions.csv`) and load it into a PostgreSQL database.
- Ensure the data is loaded efficiently, considering memory usage and performance.

2. Data transformation and aggregation

- Use *dbt* to clean and transform the raw data, creating one dimension table (`dim_table``) and one fact table (`fact_table``) based on the dataset provided.
- Transformations should address any data cleaning, and formatting.
- Standardize relevant fields, and ensure that dates, numeric fields, and identifiers are processed in a way that supports downstream use cases.
- Summarize the data to provide insights at an aggregate level, such as monthly totals or totals by customer. Consider how these summaries could be useful and what information should be included.

3. Data orchestration

- Set up *Airflow* to create a DAG that:
 - i. Downloads and ingests the raw data.
 - ii. Triggers dbt to process and transform the data into the dimensional model.
 - iii. Optionally includes task dependencies, retries, or notification logic if time allows.

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About

Please notice this role is platform focused, building frameworks and applying best practices so you should consider a high-level approach that prioritizes best practices and scalability over individual pipeline specifics. There is no need to implement everything but to demonstrate that these topics are considered.

Data quality

- Design and implement data quality checks to handle common issues, such as missing or inconsistent data. Think about what types of issues might compromise data quality and document your approach.
- Where possible, implement mechanisms to log or flag issues and address them in the data.

Other relevant topics are optimisation, automation, data governance, etc.

Technology stack

- Use Docker and `docker-compose` to manage PostgreSQL, Airflow and dbt.
- PostgreSQL for raw and processed data storage.
- dbt for data transformations and model creation.
- Airflow for orchestration.

Output - Repository structure

A well-organized GitHub repository including:

- A `README.md` with clear setup instructions and a brief description of the pipeline.
- A `docker-compose.yml` file orchestrating all services (PostgreSQL, Airflow and dbt).
- Any required dbt files, Airflow DAGs, and SQL scripts.
- The entire pipeline should be deployable with `docker-compose up`.

Submission instructions

Please complete this task before our scheduled call.

Share a GitHub repository link containing all required files and documentation replying to the email invitation please.

Please, remember there's no single right answer; we're eager to learn about your thought process and share our experiences together. Looking forward to our conversation!



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Schema for `customer_transactions.csv`

Field	Data type (expected)	Description
transaction_id	integer	Unique identifier for each transaction
customer_id	integer	Unique identifier for each customer
transaction_date	date	Date when the transaction occurred
product_id	Integer	Unique identifier for each product
product_name	String	Name of the product
quantity	Integer	Number of units sold
price	Float	Price of the product
tax	Float	Tax applied to the transaction

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